Practice Midterm

GSE 510

Total Points: 100
10 Questions, 10 Points Each

Instructions: Answer all questions. Show your work clearly. Each question is worth 10 points.

1. **Sets and Logic** **(10 points)**

Let

$$A = \{x \in \mathbb{R} : 0 \leq x \leq 10\}, \qquad B = \{x \in \mathbb{R} : x > 5\}, \qquad C = \{x \in \mathbb{R} : x < 8\}.$$

Answer the following:

(a) Find $A \cap B$.

(b) Find $A \cap C$.

(c) Find $B \cap C$.

(d) Find $A \cap B \cap C$.

(e) If the universal set is $\mathbb{R}$, find A^c.

2. **Bounds, Suprema, and Infima** **(10 points)**

Consider the set

$$S = \{x \in \mathbb{R} : 2 < x < 7\}.$$

(a) Is S bounded above?

(b) Is S bounded below?

(c) Find the least upper bound of S.

(d) Find the greatest lower bound of S.

(e) Are the least upper bound and greatest lower bound elements of S?

3. **Sequences and Limits** **(10 points)**

For each sequence, determine whether it converges. If it converges, find its limit.

(a) $x_n = 5 + \dfrac{3}{n}$

(b) $y_n = \dfrac{2n+1}{n}$

(c) $z_n = (-1)^n$

(d) $w_n = \dfrac{1}{n^2}$

4. **Open and Closed Sets** **(10 points)**

Consider the following sets:

$$A = (0,1), \qquad B = [0,1], \qquad C = (0,1].$$

(a) Is A open in $\mathbb{R}$? Is it closed?

(b) Is B open in $\mathbb{R}$? Is it closed?

(c) Is C open in $\mathbb{R}$? Is it closed?

(d) What are the boundary points of A?

(e) Give a sequence contained in A that converges to a point not contained in A.

5. **Linear Algebra and Systems of Equations** **(10 points)**

Consider the system

$$\begin{aligned} 2x_1 + x_2 &= 11, \\ x_1 + 3x_2 &= 18. \end{aligned}$$

(a) Write the system in matrix form $\mathbf{Ax} = \mathbf{b}$.

(b) Compute $|\mathbf{A}|$.

(c) Is $\mathbf{A}$ nonsingular?

(d) Solve for x_1 and x_2.

(e) Explain what nonsingularity implies about the number of solutions.

6. **Span and Linear Independence in $\mathbb{R}^2$** **(10 points)**

Let

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

(a) Write a general linear combination $a\mathbf{v}_1 + b\mathbf{v}_2$.

(b) Can $\mathbf{v}_1$ and $\mathbf{v}_2$ generate the vector

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix}?$$

If yes, find a and b.

(c) Compute the determinant of the matrix whose columns are $\mathbf{v}_1$ and $\mathbf{v}_2$.

(d) Are $\mathbf{v}_1$ and $\mathbf{v}_2$ linearly independent?

(e) Do $\mathbf{v}_1$ and $\mathbf{v}_2$ span $\mathbb{R}^2$?

7. **Linear Regression in Matrix Form** **(10 points)**

Suppose we estimate the model

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

using three observations:

$$\mathbf{y} = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}, \qquad \mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}.$$

(a) Write the regression model in matrix form.

(b) Compute $\mathbf{X}'\mathbf{X}$.

(c) Compute $\mathbf{X}'\mathbf{y}$.

(d) Compute

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

(e) Write the fitted regression line.

8. **Differentiation and Economic Interpretation** **(10 points)**

Suppose a firm's total cost function is

$$C(q) = 50 + 10q + q^2.$$

(a) Find the marginal cost function $C'(q)$.

(b) Find $C'(20)$.

(c) Find $C''(q)$.

(d) Is marginal cost increasing or decreasing in q?

(e) Use a linear approximation to estimate the increase in cost when output rises from 20 to 21.

9. **Approximation** **(10 points)**

Suppose output is produced according to

$$Y = F(K) = 10K^{1/2}.$$

The current level of capital is $K_0 = 100$.

(a) Find $F'(K)$.

(b) Find $F''(K)$.

(c) Use the linear approximation around $K_0 = 100$ to approximate $F(104)$.

(d) Use the quadratic approximation

$$F(K) \approx F(100) + F'(100)(K - 100) + \frac{1}{2}F''(100)(K - 100)^2$$

to approximate $F(104)$.

(e) Which approximation would you expect to be closer to the true value? Briefly explain.

10. **Implicit Differentiation and Comparative Statics** **(10 points)**

Suppose market equilibrium is defined by

$$D(p, a) = S(p),$$

where demand is

$$D(p, a) = a - 2p$$

and supply is

$$S(p) = 10 + p.$$

Here a is a demand-shift parameter.

(a) Rewrite the equilibrium condition as

$$F(p, a) = 0.$$

(b) Solve explicitly for the equilibrium price $p^*(a)$.

(c) Compute $\frac{dp^*}{da}$ using your explicit solution.

(d) Use the implicit function theorem formula

$$\frac{dp^*}{da} = -\frac{F_a}{F_p}$$

to verify your answer.

(e) Interpret $\frac{dp^*}{da}$ economically.